

Customer Churn Prediction using Machine Learning

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Python Libraries used:

- Numpy
- Pandas
- Matplotlib
- Scikit-learn

Project Objective

The objective of this project was to develop a machine learning model capable of predicting whether a telecom customer is likely to churn (leave the service). The project focused on building a complete machine learning pipeline, comparing multiple classification algorithms, and identifying the most influential factors affecting customer churn.

Dataset

The project used a telecom customer churn dataset containing demographic information, subscribed services, billing details, contract type, payment methods, and the target variable 'Churn'. The dataset contains approximately **7,000 telecom customer records** with 20 input features and one target variable.

Data Preprocessing

Missing values in the TotalCharges column were identified and removed. The customerID column was excluded since it does not contribute to prediction. Categorical variables were converted into numerical form using One-Hot Encoding, and numerical features were standardized using StandardScaler.

Exploratory Data Analysis

Exploratory analysis was carried out to understand customer behaviour and the relationship between different features and churn. Features such as Contract, InternetService, OnlineSecurity, TechSupport, and PaymentMethod showed strong relationships with churn.

Model Development

Five classification algorithms were trained and evaluated: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN). Data was divided into training (80%) and testing (20%) sets.

Model Evaluation

Models were compared using Accuracy, Precision, Recall, and F1-score. Logistic Regression achieved the best overall performance among the baseline models with approximately 80.5% accuracy and an F1-score of about 0.61.

Hyperparameter Tuning

Random Forest was further optimized using GridSearchCV with 5-fold cross-validation. The tuned model achieved performance similar to the default model, indicating that extensive tuning did not significantly improve prediction for this dataset.

Feature Importance

Feature importance scores obtained from the Random Forest model were used to rank the predictors. The ten most important features were selected to investigate whether a simpler model could achieve comparable performance.

Feature Selection

Logistic Regression was retrained using only the top ten important features. Although the simplified model experienced a slight reduction in performance, it demonstrated that a smaller feature set could still provide competitive results while improving interpretability.

Conclusion

This project demonstrated a complete machine learning workflow, including data preprocessing, exploratory analysis, model development, evaluation, hyperparameter tuning, and feature selection. Logistic Regression emerged as the most effective model for this dataset. The project also highlighted the trade-off between model simplicity and predictive performance when reducing the number of features.